DP024
Physics 2
Semester 2
Session 2024/2025
2 hours

UNIT FIZIK
KOLEJ MATRIKULASI KELANTAN

PRA-PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI 4
MATRICULATION PROGRAMME PRE-EXAMINATION 4

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU
NO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

NAME: ____________________________

PRAC: ____________________________

MATRIC NO: ______________________

NO	Full Marks	Mark(s)
1		
2		
3		
4		
5		
6		
Total		

Kertas soalan ini mengandungi 9 halaman bercetak.
This question paper consists of 9 printed pages.

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum *Laju cahaya dalam vakum*	c	$= 3.00 \times 10^{8}$ m s^{-1}
Permeability of free space *Ketelapan ruang bebas*	μ_o	$= 4\pi \times 10^{-7}$ H m^{-1}
Permittivity of free space *Ketelusan ruang bebas*	ε_o	$= 8.85 \times 10^{-12}$ F m^{-1}
Electron charge magnitude *Magnitud cas elektron*	e	$= 1.60 \times 10^{-19}$ C
Planck constant *Pemalar Planck*	h	$= 6.63 \times 10^{-34}$ J s
Electron mass *Jisim elektron*	m_e	$= 9.11 \times 10^{-31}$ kg $= 5.49 \times 10^{-4}$ u
Neutron mass *Jisim neutron*	m_n	$= 1.674 \times 10^{-27}$ kg $= 1.008665$ u
Proton mass *Jisim proton*	m_p	$= 1.672 \times 10^{-27}$ kg $= 1.007277$ u
Deuteron mass *Jisim deuteron*	m_d	$= 3.34 \times 10^{-27}$ kg $= 2.014102$ u
Molar gas constant *Pemalar gas molar*	R	$= 8.31$ J K^{-1} mol^{-1}
Avogadro constant *Pemalar Avogadro*	N_A	$= 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant *Pemalar Boltzmann*	k	$= 1.38 \times 10^{-23}$ J K^{-1}
Free-fall acceleration *Pecutan jatuh bebas*	g	$= 9.81$ m s^{-2}

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Atomic mass unit *Unit jisim atom*	1 u	$= 1.66 \times 10^{-27}$ kg $= 931.5 \ \frac{MeV}{c^2}$
Electron volt *Elektron volt*	1 eV	$= 1.6 \times 10^{-19}$ J
Constant of proportionality for Coulomb's law *Pemalar hukum Coulomb*	$k = \frac{1}{4\pi\varepsilon_\circ}$	$= 9.0 \times 10^{9}$ N m^{2} C^{-2}
Atmospheric pressure *Tekanan atmosfera*	1 atm	$= 1.013 \times 10^{5}$ Pa
Density of water *Ketumpatan air*	ρ_w	$= 1000$ kg m^{-3}

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

1. $x = A\sin\omega t$
2. $\omega = \frac{2\pi}{T} = 2\pi f$
3. $v = A\omega \cos \omega t$
4. $a = -A\omega^2 \sin \omega t$
5. $k = \frac{2\pi}{\lambda}$
6. $v = f\lambda$
7. $y(x,t) = A\sin(\omega t \pm kx)$
8. $v = \frac{dy}{dt}$
9. $y = 2A \cos kx \sin \omega t$
10. $v = \sqrt{\frac{T}{\mu}}$
11. $I = \frac{dQ}{dt}$
12. $Q = ne$
13. $\rho = \frac{RA}{l}$
14. $R_{eff} = R_1 + R_2 + R_3 + \cdots + R_n$
15. $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$
16. $V = IR$
17. $C = \frac{Q}{V}$
18. $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_n}$
19. $C_{eff} = C_1 + C_2 + C_3 + \cdots + C_n$
20. $\tau = RC$
21. $Q = Q_o e^{\frac{-t}{RC}}$
22. $Q = Q_o\left(1 - e^{\frac{-t}{RC}}\right)$
23. $B = \frac{\mu_o I}{2\pi r}$
24. $B = \frac{\mu_o NI}{L}$
25. $B = \frac{\mu_o I}{2r}$
26. $\phi = BA\cos\theta$
27. $\Phi = N\phi$
28. $\varepsilon = -\frac{d\phi}{dt}$
29. $\varepsilon = Blv\sin\theta$
30. $\varepsilon = -NA\frac{dB}{dt}$
31. $\varepsilon = -NB\frac{dA}{dt}$
32. $R = 2f$

33. $$\frac{1}{f}=\frac{1}{u}+\frac{1}{v}$$

34. $$m=\frac{h_i}{h_o}=-\frac{v}{u}$$

35. $$y_m=\frac{m\lambda D}{d}$$

36. $$y_m=\frac{\left(m+\frac{1}{2}\right)\lambda D}{d}$$

37. $$\Delta y=\frac{\lambda D}{d}$$

Answer **all** questions.

1. (a) A wave represented by an equation $\boldsymbol{y(x,t) = 0.4\sin\,(pt - 6x)}$ where x and y are in meters and t is in second. The speed of the wave is 0.5 m s^{-1}. Determine the

(i) amplitude and direction of wave propagate.

(ii) wave number.

(iii) distance between crest to crest.

(iv) value of $\boldsymbol{p}$ and name the parameter.

[8 *marks*]

(b) A standing wave on a stretched string is described by the equation:

$$y(x,t) = 0.8\cos\left(\frac{\pi}{5}x\right) sin(100\pi t)$$

where x and y are in centimeters and t is in second. Calculate the

(i) speed of the wave.

(ii) amplitude of the wave at the point 0.25 cm from the origin.

(iii) tension in the string if the string has a mass of 3.0 g and length of 1.5 m.

[8 *marks*]

2. (a) A silver wire has resistance of 0.25 Ω and a cross-sectional area of 0.80 mm^2. The resistivity of silver wire is $1.59 \times 10^{-9}\,\Omega$ m. Calculate the length of the wire.

[2 *marks*]

(b) A current of 1.5 A flows through a wire for 10 s. Calculate the number of electrons that pass through a cross-section of wire in that time.

[2 *marks*]

(c)

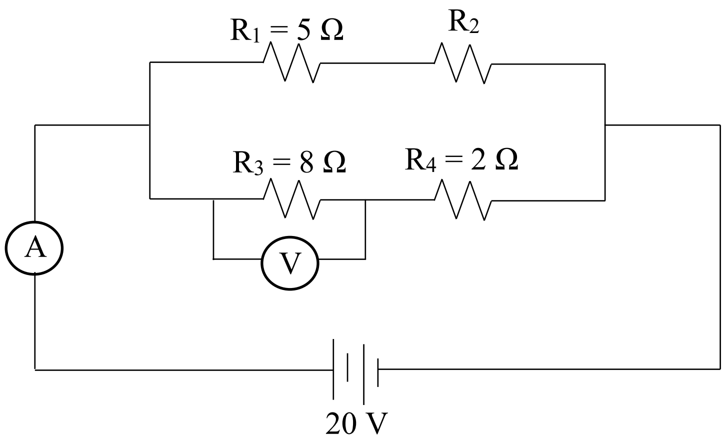

FIGURE 1

FIGURE 1 shows a circuit consisting of four resistors connected to 20 V power supply. The reading of the ammeter is 4 A.

(i) Calculate the value of R_2.

(ii) What is the voltmeter reading.

[8 *marks*]

3. (a) A capacitor has a charge of 150 nC when the potential difference across it is 3.0 V.

(i) Calculate the capacitance of the capacitor.

(ii) Determine the charge stored on the capacitor when the potential difference is increased to 4.7 V.

[3 *marks*]

(b)

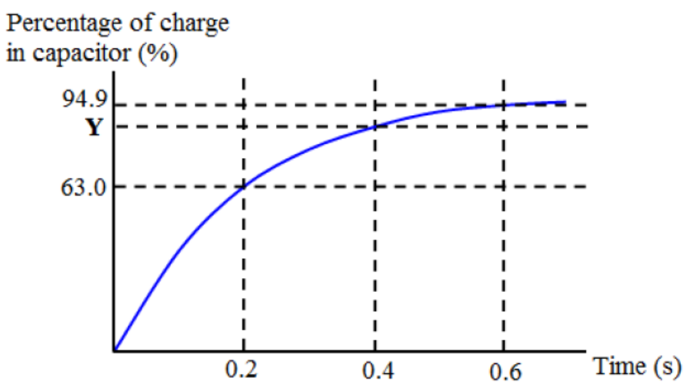

FIGURE 2

FIGURE 2 above represent the percentage of charge variation in a capacitor over time when it is connected in series to a 5 Ω resistor and a 10 V power supply. Determine the

(i) time constant.

(ii) capacitance of capacitor.

(iii) amount of charge in the capacitor at $t = 0.6$ s.

(iv) percentage of charge in the capacitor at Y.

[7 *marks*]

4. (a) A metal rod moves perpendicularly through a 1.5 T uniform magnetic field at a speed of 2 cm s^{-1}. If the length of the rod is 40 cm and its resistance is 3 Ω, calculate

(i) the induced emf.

(ii) the induced current.

[4 *marks*]

(b) A 400 turns circular coil has a cross-sectional area 1.81×10^{-3} m^2 and carries a current of 3.4 A. The coil is perpendicular to the magnetic field. Determine the

(i) magnitude of magnetic field at the centre of the coil.

(ii) magnetic flux through the coil.

(iii) magnetic flux linkage through the coil.

(iv) magnitude of the induced emf in the coil in 1 minute if the coil is rotated so that it makes an angle of 20° to the magnetic field.

[10 *marks*]

5. (a) A concave mirror has radius of curvature 30 cm. An object height 12 cm forms an image 20 cm at the back of the mirror.

(i) Calculate the object distance.

(ii) Calculate the height of the image.

(iii) State the characteristics of the image.

[7 *marks*]

(b) An object of 15 cm high is placed at 0.10 m in front of diverging lens. If the focal length is 0.25 m,

(i) calculate the image position.

(ii) calculate the magnification of the lens.

(iii) sketch the ray diagram of the diverging lens.

[7 *marks*]

6. (a) A double-slits pattern is view on a screen 1.5 m from the slit. If the third order minima are 20 cm apart, determine the ratio of wavelength and separation between the slits.

[3 *marks*]

(b) The separation of the double slits in a Young's two-slit experiment is 0.4 mm and the distance of the screen to the slits is 2.0 m. If the wavelength of the light used is 500 nm, determine

(i) the fringe separation

(ii) the distance between of 5^{th} bright fringe from the central maximum.

[5 *marks*]

(c) In Young's double-slits experiment, the slits are 0.3 mm apart and the distance of the screen to the slits is 50 cm. The slits are illuminated by an orange light of wavelength 6.0×10^{-7} m.

(i) Calculate the separation between the adjacent dark fringes.

(ii) What will happen to the interference pattern if a light of shorter wavelength is used instead? Explain your answer.

(iii) What is observed on the screen if one of the slits is covered? Explain your answer.

[6 *marks*]

END OF QUESTION PAPER

KERTAS SOALAN TAMAT